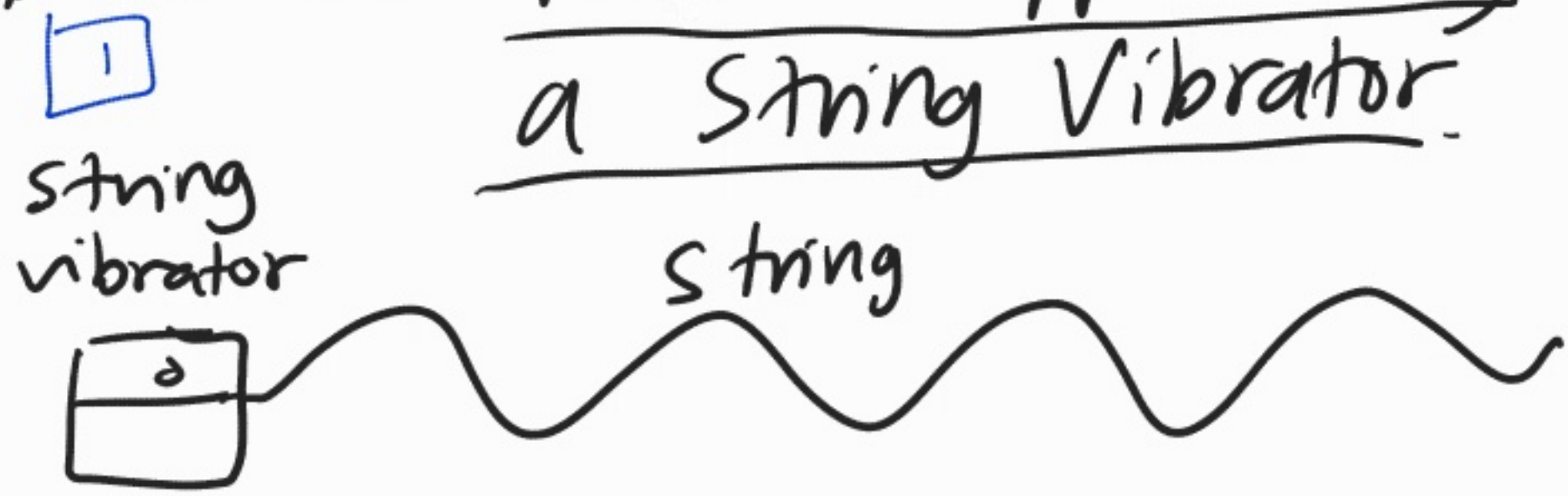
Exercise: Power Supplied by a String Vibrator



A Sinusoidal wave on a string is produced by a string vibrator. The string has a $m = 80g$, $\tau = 90N$ and a length of $l = 2.5m$. It oscillates with a $f = 60\,Hz$ and an amplitude of $5\,cm$. Considering the wave travels at constant speed, determine the time-averaged power supplied to the wave by the string vibrator.

Solution :

⇒ Time-averaged power

$$P = \frac{1}{2} \mu y_m^2 \omega^2 v$$

1, Calculate μ to find the wave speed :

$$\mu = \frac{m}{L} = \frac{0.08\,kg}{2.5\,m} = 0.032\,kg/m$$

2, Determine wave speed :

$$V = \sqrt{\frac{\tau}{\mu}} = \sqrt{\frac{90N}{0.032\,kg/m}} = 53.03\ m/s$$

3, Calculate angular frequency ω.

$$\omega = 2\pi f = 2\pi(60\,Hz)$$
$$= 376.8\,s^{-1}$$

4, Determine Time-Averaged Power

$$P = \frac{1}{2}\mu y_m^2 \omega^2 v$$

$$= \left[\frac{1}{2}(0.5)(0.032)^2(0.05)(376.8)^2\right] W$$
$$\times\ 53.03$$

$$= 301.16\,W$$

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Exercise 2

Consider a sound wave with the power of 5 μW. If the source of this sound wave is a point source:

a) Determine the intensity of this sound wave from a distance of 5m from the source.

b) Calculate the sound level at that distance.

Solution:

a) Point Source → Spherical propagation.

$$I = \frac{5\times10^{-6}\ W}{4\pi(5m)^2} \approx \underline{1.592\times10^{-8}\ W/m^2}$$

b) Sound Level:

$$\beta = 10\log_{10}\left[\frac{I}{I_0}\right]$$

$$I_0 = 1\times10^{-12}\ W/m^2$$

$$\therefore \beta = 10\log_{10}\left[\frac{1.592\times10^{-8}\ W/m^2}{1\times10^{-12}\ W/m^2}\right]$$

$$\approx \underline{42\ dB}$$

Exercise 3

A ten-minute exposure of 120dB would shift your hearing threshold at 1000 Hz from 0dB to 28dB. Ten years exposure to 92dB causes a permanent shift of your hearing threshold to 28dB. What is the ratio of sound intensities between both exposure scenarios? Discuss your conclusion from this result.

Solution.

For the 10-minute exposure:

$$I_{120} = I_0 \, 10^{\beta/10\,dB}$$

$$= \left(10^{-12}\ W/m^2\right)(10)^{12} = 1\ W/m^2$$

For the 10-year exposure:

$$I_{92} = \left(10^{-12}\ W/m^2\right)(10)^{9.2} = 10^{-2.8}\ W/m^2$$

$$\frac{I_{92}}{I_{120}} = 10^{-2.8}$$

⇒ Although the 10-year exposure scenario has a weaker sound intensity by a factor $10^{-2.8}$, the hearing damage is more serious. Thus exposure time frames play a critical role in hearing safety.